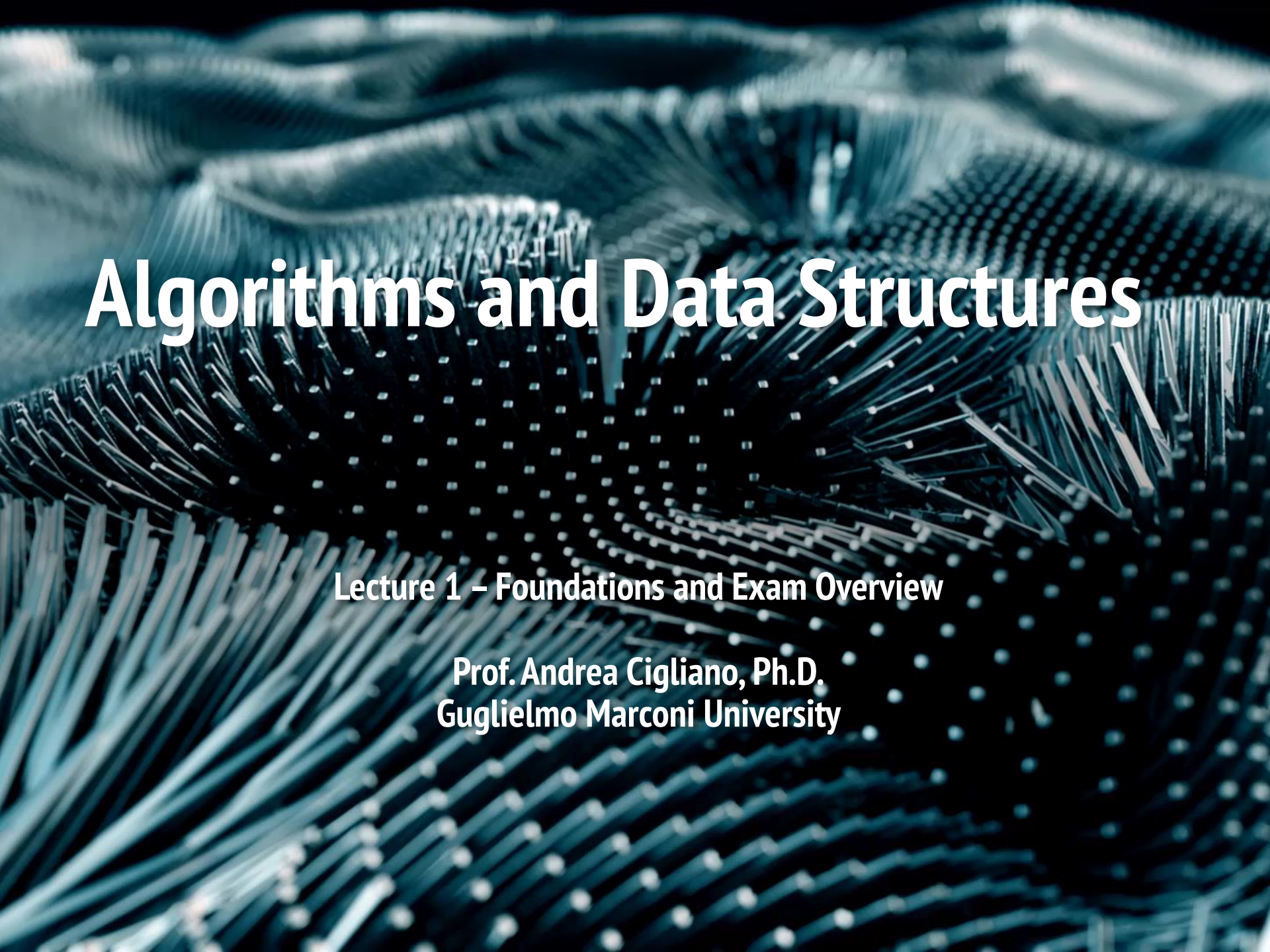
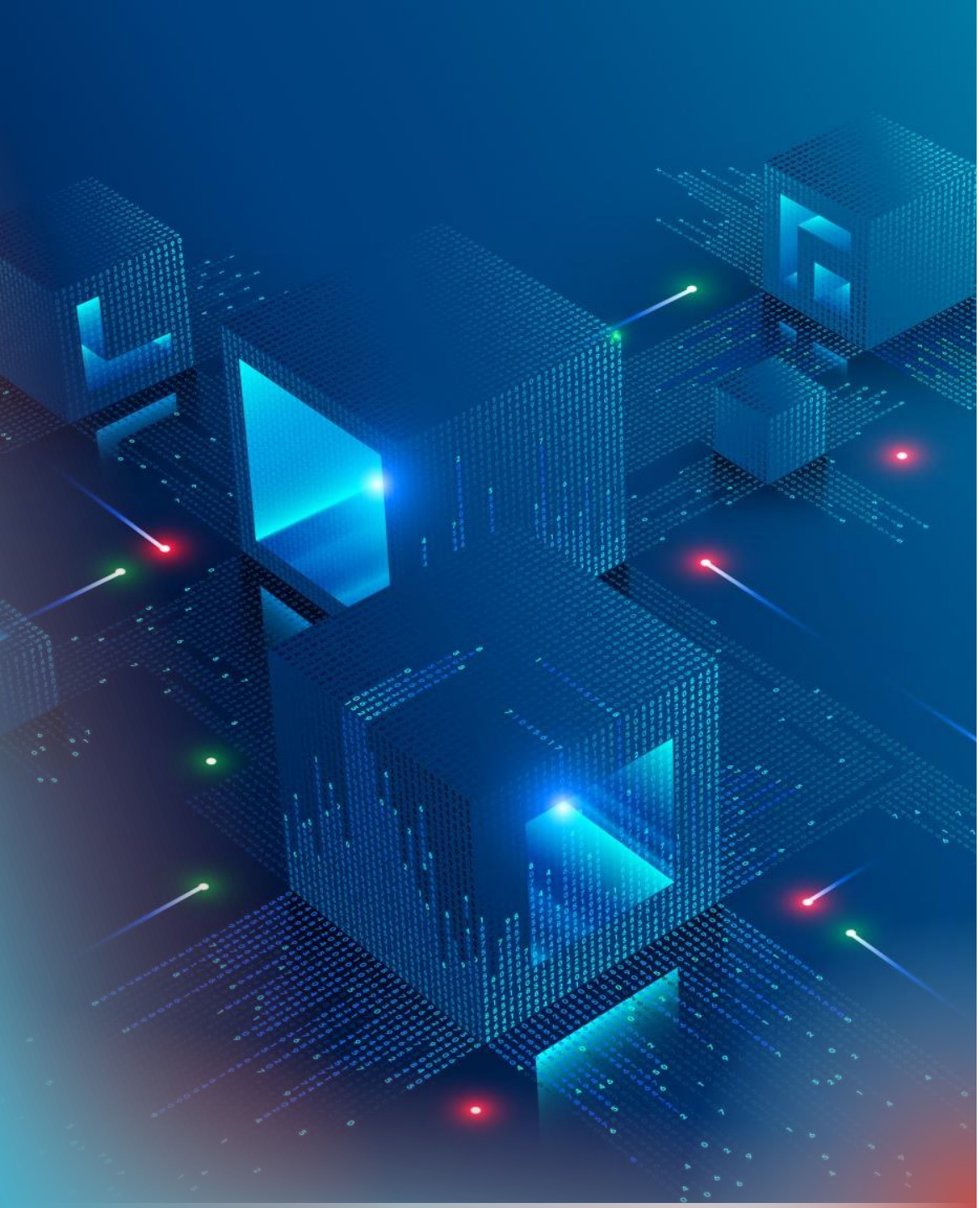


Algorithms and Data Structures

Lecture 1 – Foundations and Exam Overview

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Course Overview

The course in Algorithms and Data Structures provides students with the fundamental foundations for designing and analyzing efficient algorithms and choosing appropriate data structures.



Course Goals

Knowledge and understanding:

- Understand the fundamental concepts of data structures and algorithms
- Know the main techniques for algorithm design
- Understand methods for analyzing computational complexity

Applying knowledge and understanding:

- Be able to choose and implement appropriate data structures for specific problems
- Apply algorithm design techniques to solve computational problems
- Analyze algorithm efficiency in terms of time and space



Transversal skills

- Making judgements
  - Critically evaluate the effectiveness and efficiency of different algorithmic solutions
  - Choose the most suitable algorithm or data structure based on problem specifications
  - Understand trade-offs between different algorithmic solutions
- Communication skills
  - Clearly describe algorithms and data structures using pseudocode and diagrams
  - Present and justify algorithmic choices clearly and convincingly
  - Effectively communicate complex concepts related to algorithms and data structures
- Learning skills
  - Develop the ability to learn new algorithms and data structures independently
  - Acquire the foundation to tackle advanced courses and more complex problems in computer engineering
  - Develop an analytical and creative approach to solving computational problems

These objectives aim to provide students with not only theoretical knowledge but also practical skills and analytical abilities essential for their future careers in computer engineering and software development.



Course requirements

To effectively achieve the learning objectives of the Algorithms and Data Structures course, the following foundational knowledge is recommended:

Programming

- Basic knowledge of at least one programming language (typically Python)
- Understanding of fundamental programming concepts such as variables, loops, functions, and control structures

- Discrete Mathematics
  - Fundamentals of logic and set theory
  - Knowledge of mathematical proof techniques
  - Understanding of concepts such as relations, functions, and graphs.
- Calculus
  - Basic knowledge of differential and integral calculus
  - Understanding of series and sequences
- Linear Algebra
  - Basic knowledge of vectors and matrices
  - Understanding of vector space concepts
- Computer Architecture
  - Basic understanding of computer architecture
  - Knowledge of number systems and data representation
- Foundations of Computer Science
  - Knowledge of theoretical computer science fundamentals
  - Understanding of the concept of algorithm and computability
- Probability and Statistics (for some advanced topics)
  - Basic knowledge of probability
  - Understanding of fundamental statistical concepts



Course requirements

This foundational knowledge ensures that students are prepared to:

- Understand and implement complex data structures
- Mathematically analyze algorithm efficiency
- Apply algorithm design techniques to real-world problems
- Understand the practical implications of algorithmic choices in terms of computational performance



Course requirements

MODULE 1 - Introduction to algorithms
Introduction to data structures and algorithms

- Sorting algorithms – Part I
- Sorting algorithms – Part II
- Optimization algorithms and graphs
- Analysis methods for recursive algorithms

MODULE 2 - Introduction to data structures
Data structures – Part I

- Data structures – Part II
- Data structures – Part III

MODULE 3 - Applications and theory
Greedy algorithms

- Flow networks
- Theory of np-completeness

MODULE 4 - Advanced techniques
Advanced design and analysis techniques

- Advanced data structures

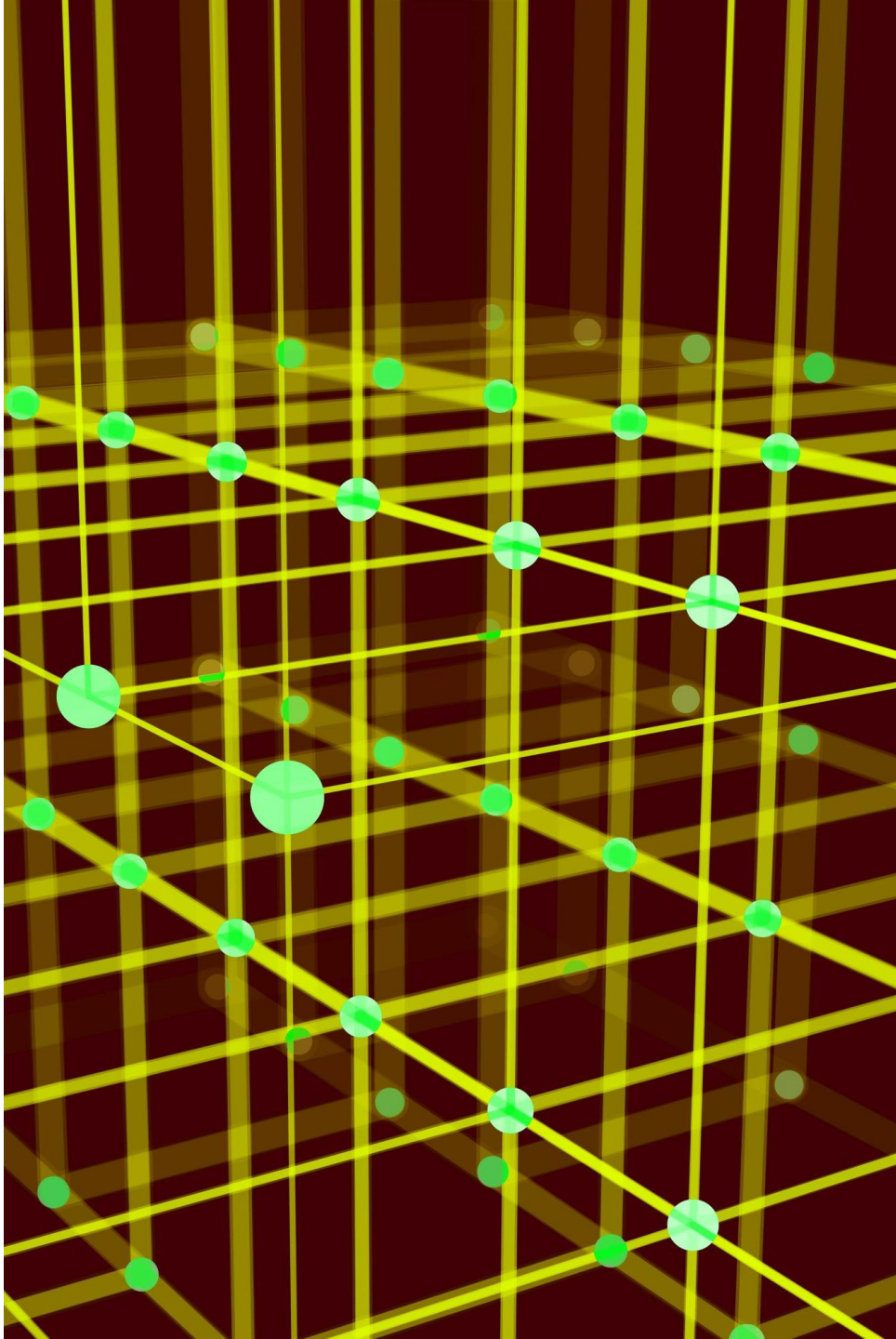
MODULE 5 Algorithmic insights
Machine learning algorithms

- Parallel and online algorithms
- Number theory algorithms
- String matching algorithms
- Approximation algorithms
- Genetic algorithms
- Quantum algorithms

Algorithms and Data Structures

Exam Structure

- The exam is designed to assess both theoretical understanding and practical problem-solving skills in Algorithms and Data Structures.
- The exam is written. It consists 2 parts, the first in 4 open-ended questions and the second in 5 multiple-choice questions, with a maximum duration of two hours.
- In particular:



Algorithms and Data Structures

Exam Structure

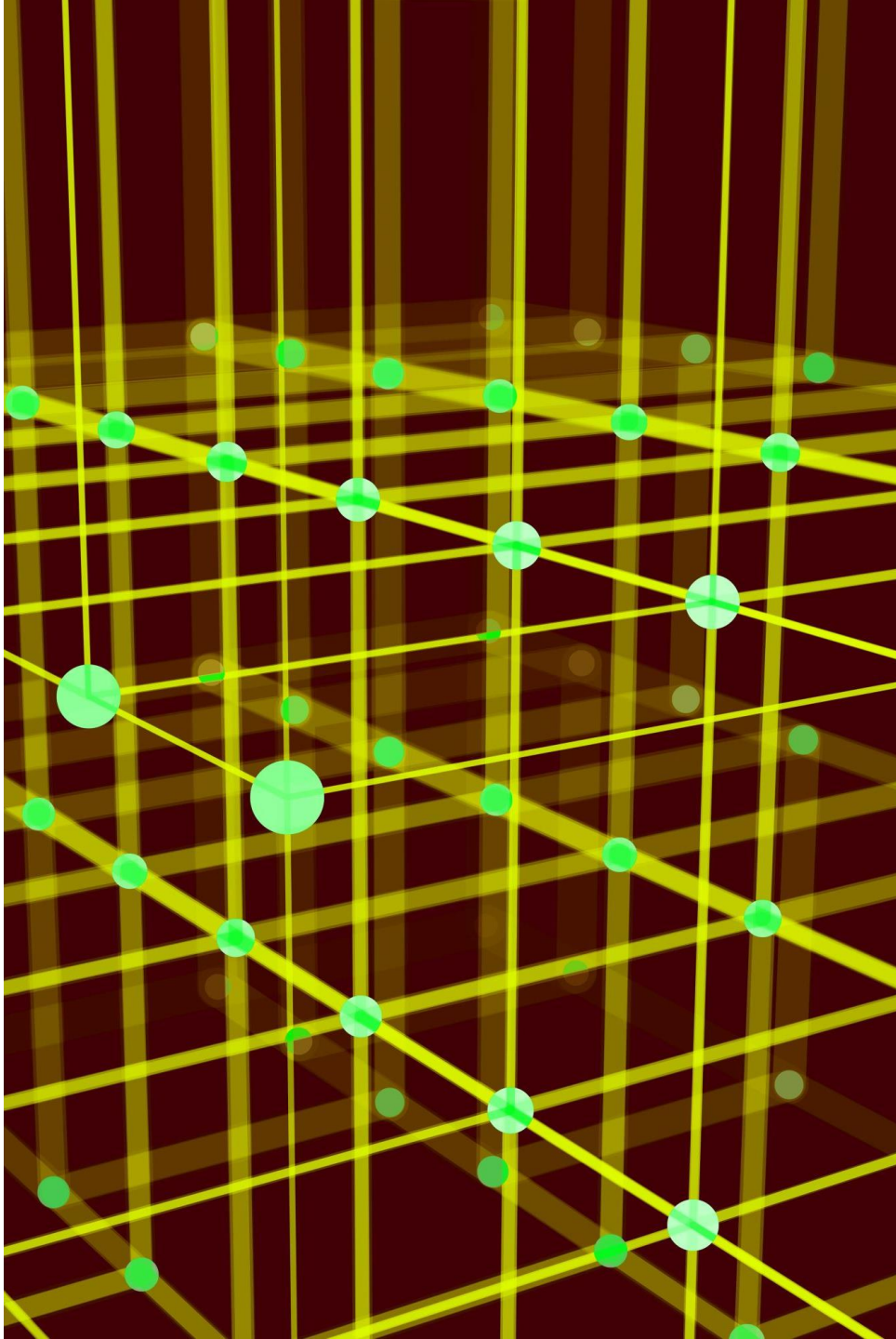
1. Open-Ended Questions

The exam includes 4 open-ended questions.
These questions require detailed, structured answers in English.
Students are expected to:

- Explain key theoretical concepts (e.g., asymptotic complexity, data structures)
- Describe and analyze algorithms
- Justify design choices using time and space complexity
- Apply knowledge to realistic scenarios or case studies
- Typical topics include:
- Complexity analysis (worst, average, best case)
- Balanced trees (e.g., red-black trees)
- Graph algorithms (e.g., max flow, shortest paths)
- Design of efficient systems using appropriate data structures

Answers should demonstrate:

- Clarity and precision
- Correct use of terminology
- Ability to connect theory with practice



Algorithms and Data Structures

Exam Structure

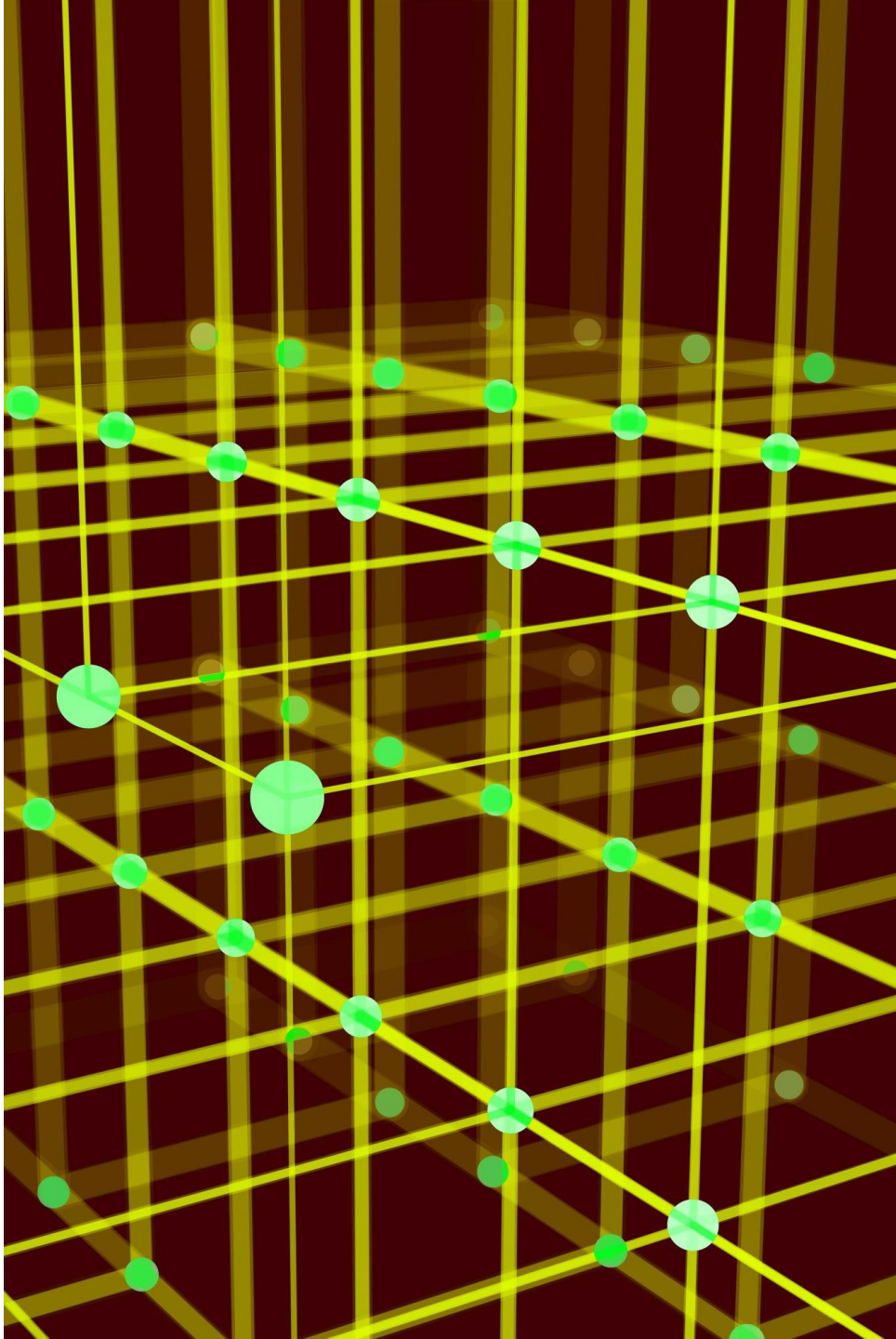
2. Multiple-Choice Questions

The exam includes 5 multiple-choice questions.
Each question has four possible answers, with only one correct option.
These questions are designed to test:

- Core definitions and concepts
- Knowledge of algorithmic complexity
- Recognition of correct algorithm–problem matches
- Understanding of fundamental data structures

Example main topics:

- Time complexity of sorting algorithms
- Priority queue implementations
- Shortest path algorithms
- NP-completeness
- String matching algorithms



Algorithms and Data Structures

Exam Structure

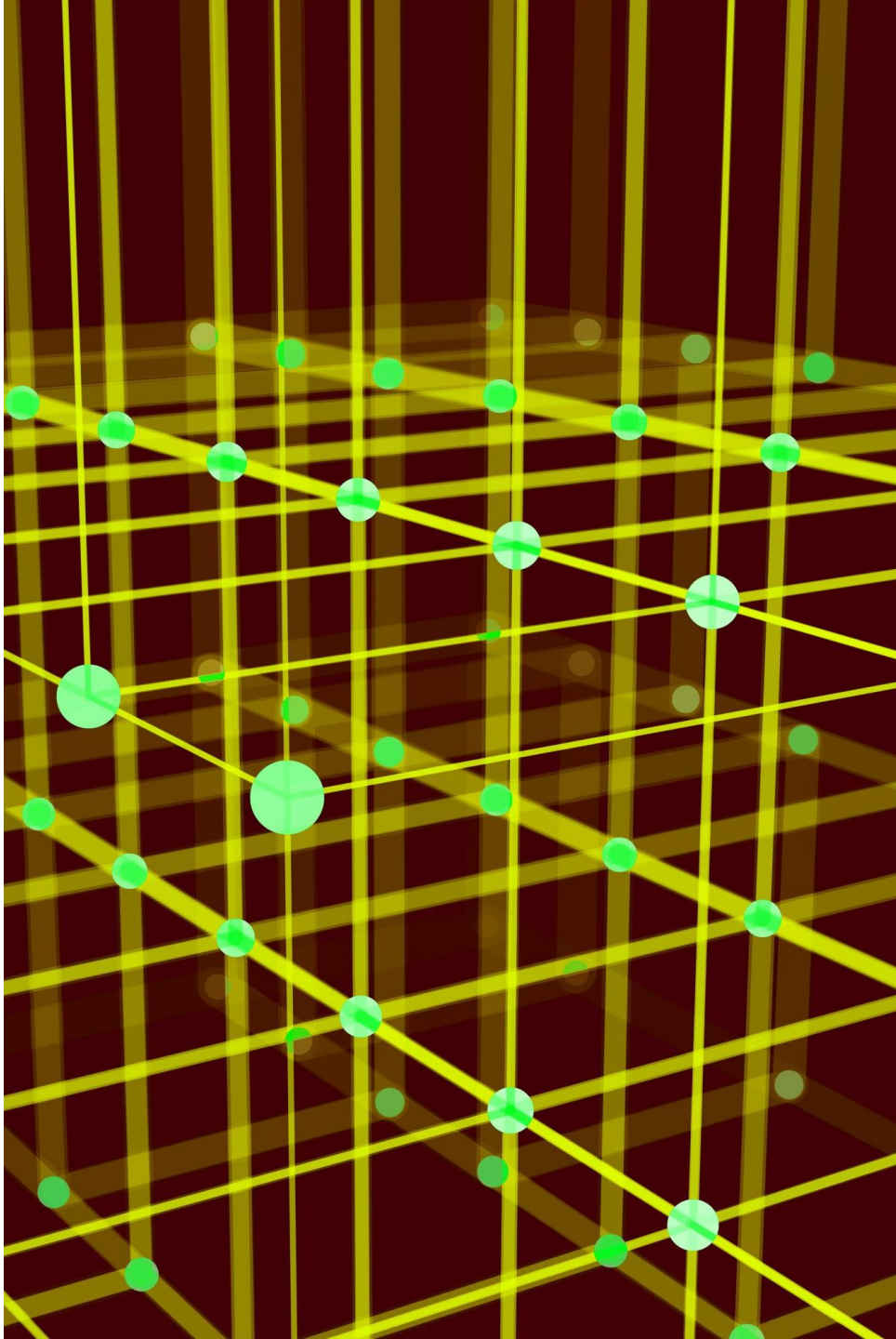
Evaluation Criteria

The final assessment is based on:

- Correctness and completeness of open-ended answers
- Quality of reasoning and justification
- Accuracy in multiple-choice responses
- Answers enriched with code snippets (e.g. Python, pseudolanguage, Java, etc...)

Particular attention is given to:

- Proper use of asymptotic notation
- Ability to select appropriate data structures
- Understanding of algorithmic efficiency

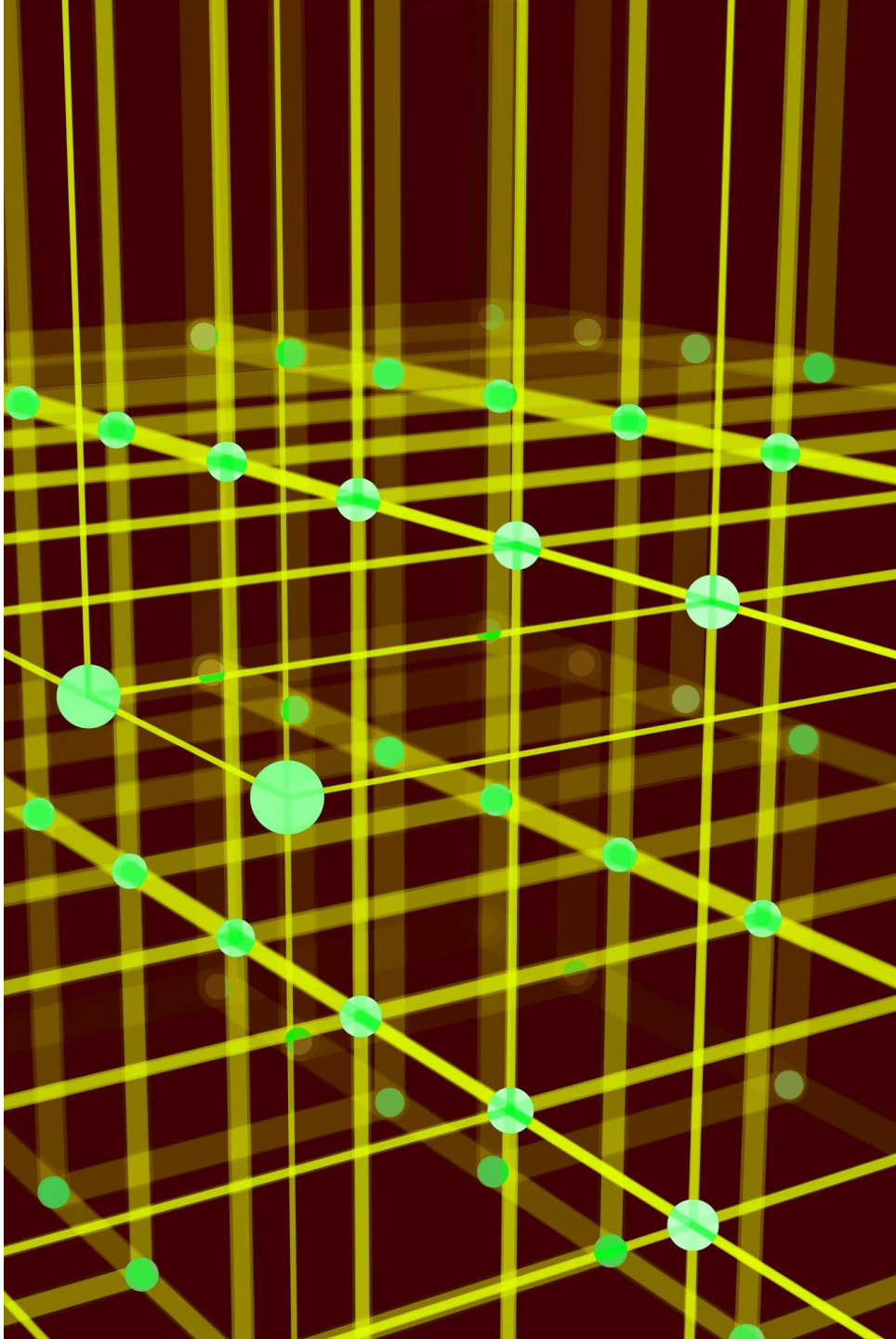


Algorithms and Data Structures

Exam Structure

Language Requirement

- The exam must be answered in English
- Students should use formal and technical language, as expected in an academic context



Unimarconi

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